europass



Zeynep Sevgi Çelik

Date of birth: 23/07/1995 | **Place of birth:** Istanbul, Türkiye | **Nationality:** Turkish | **Phone:** (+90) 5345459515 (Mobile) | **Email address:** z.sevgicelik@gmail.com | **Website:** zeynepsevgicelik.github.io | **LinkedIn:** linkedin.com/in/zeynepsevgicelik | **Address:** Üsküdar, 34676, Istanbul, Türkiye (Home)

ABOUT ME

Innovative Mechatronics Engineer with practical experience in C programming, firmware development, embedded systems, electronic circuit design, and microcontroller programming. Skilled in working with various sensors, automation, control systems, and safety-critical applications. Strong problem-solving, teamwork, and communication skills; passionate about developing reliable and efficient engineering solutions. Currently expanding skills and proficiency in C++ development.

WORK EXPERIENCE

13/11/2020 - 14/01/2025 - ISTANBUL, TÜRKIYE

EMBEDDED SYSTEMS SOFTWARE DEVELOPER FERVES TECHNOLOGY

- Developed firmware in C for microcontroller-based systems.
- Designed AC/DC circuits and integrated sensors for prototypes.
- Contributed to R&D and product improvement projects.
- Led project team and participated in on-site/field testing of hardware and firmware.
- Supported production, quality control, and testing teams.
- Conducted research and cost analysis for component selection.

Business or sector Manufacturing | **Department:** Research and Development | **Website:** https://ferves.com/

22/01/2020 - 16/08/2020 - ISTANBUL, TÜRKIYE

SERVER ADMINISTRATOR DORABASE - VUCOS

- Managed VoD content deployment and scheduling for Vodafone TV.
- Administered live broadcast traffic and ensured smooth streaming.
- Configured servers and virtual machines to support content delivery.

Website: https://www.dorabase.com/

14/10/2018 - 09/01/2019 - AVEIRO, PORTUGAL

MECHATRONICS ENGINEER AZ MECATRÓNICA LDA

- Completed internship involving the design and testing of a pH meter circuit for sensor-based measurement applications.

15/10/2017 - 15/03/2018 - ISTANBUL, TÜRKIYE

AUTOMATION ENGINEER OTOMATICA AUTOMATION CONTROL AND SOFTWARE SERVICE

- Completed internship tasks performing BACnet alarm system revisions and designing TGML interfaces for building automation projects.

Website: https://otomatica.com/

07/08/2017 - 07/09/2017 - ISTANBUL, TÜRKIYE

MECHATRONICS ENGINEER ARÇELIK INC.

- Utilized encoders and Siemens PLCs, and designed HMI interfaces for automated system applications.
- Internship role assisting in the implementation of an image processing system to monitor drum position stability on the washing machine production line.

EDUCATION & TRAINING

16/09/2013 - 18/05/2018 - SAKARYA, TÜRKIYE

BACHELOR OF SCIENCE (B.SC.) IN MECHATRONICS ENGINEERING - SAKARYA UNIVERSITY

Field(s) of study: Electronics and automation | **Level in EQF:** 6 | **National classification:** Recognized under European Accreditation (EQF/EHEA)

14/09/2009 - 14/08/2013 - ISTANBUL, TÜRKIYE

HIGH SCHOOL DIPLOMA (ANATOLIAN HIGH SCHOOL) - KANDILLI ANATOLIAN HIGH SCHOOL

Field(s) of study: Natural sciences, mathematics and statistics | **Level in EQF:** 4

LANGUAGE SKILLS

Mother tongue(s): **TURKISH**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

PROJECTS

Vehicle Control Systems Single Motor & Trigger Boards – SW & HW, position control, adjustable speed, dual-mode operation, digital current control, USART.

Multi-output Motor Board – HW, migration from 8-bit to 32-bit processor.

Dimmer Board – SW, multitasking, cycle mode, adjustable fading, LED high-current output, variable value memorization.

WIFI Module – SW, encryption update, country-specific firmware.

Hygiene Systems: Drain Pump, Smart Pump, Diffuser Board – HW & SW, AC/Rechargeable, ESP8266, RTC, GSM, shift scheduling, WIFI, 5 operational modes.

Awning Systems Economical Series Awning – SW, tubular & 24VDC motors, thermal & digital current control, USART, RF, SPI, memorization.

Exxen Awning Control – HW & SW, AC supply, RF/WIFI, RGB multifunction, memorization.

Axis Awning Control – HW & SW, 24VDC, WIFI, LoRa, encoder-based synchronous motor driving, RGB multifunction, digital current control, memorization.

Vertex Zip & Guillotine Glass – Wind & Rain Sensors, Smart Glass, High-tech controls.

Heater Systems Dry Contact Control – HW & SW, AC supply, RF communication.

External Projects Smart Table – SW, encoder, synchronized WIFI, distance memory, soft speed control, UART.

Water Purification Board – SW, filter life memory, flowmeter, LCD, pressure & leak control.

Server Room Sensor Board – HW & SW, custom-designed Transformer-Rectifier and DC-DC Buck Converter circuits, multi-sensor interface (humidity, CO, microphone, LDR, motion), Ai-WB2-12F WiFi/Bluetooth, periodic

data acquisition and server transmission.

SKILLS

C++ | Microcontroller | ARM | STM32 | integrated development environment software | firmware | Gitlab | embedded systems | keil uVision | IAR workbench | hardware testing methods | in-circuit test | printed circuit boards testing methods | computer programming | project management | hardware components | microprocessors | sensors | ICT communications protocols | UART | USART | I2C | SPI | EEPROM | work in teams | Timers | PWM | Interrupts | GPIO | ADC | DAC | processor hardware peripherals | Visual Studio | ICT debugging tools | FreeRTOS | Embedded C | Arduino | programmable logic controller | transmission technology | test wireless devices | calculate production costs | integrate system components | semiconductors | design integrated circuits | LED lighting components | electronics | Geehy | Xcode | Bare Metal programming | Flash memory programming | NVM | Encoder | GSM | LoRa | RF | Bluetooth | esp8266 | sim800c | Wi-Fi | PIC | GITHub

CERTIFICATIONS

09/02/2025 PLEPA
C++ Programming Language Course

Completed a 220-hour program under the instruction of Necati Ergin, focusing on object-oriented programming, software architecture, and modern C++ development practices.

05/02/2023 CSD
C Programming Language Course

Completed a comprehensive 220-hour program under the instruction of Necati Ergin, covering structured programming, memory management, and embedded system fundamentals.

06/01/2019 Ayvansaray University
System and Network Administration

DRIVING LICENCE

Driving licence: B

VOLUNTEERING

14/01/2025 - 09/09/2025 Istanbul
Preply Provided free Turkish language lessons multiple times, both through Preply and online sessions, as a volunteer.

HOBBIES AND INTERESTS

Dancing

Singing

REFERENCES

Ömer Faruk Gözderesi - Senior Embedded Systems Engineer

ENDA, Istanbul / +90 543 334 38 34

Ceyhun İbolar - Senior Software Engineer

AVL, Istanbul / +90 537 467 85 77